

Reduction to Fargues - Scholze

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$\mathcal{Z}_R(G)$ Bernstein centre of $\text{Rep}_R(G)$

$$\mathcal{Z}_R(G) \longrightarrow \text{End}_{R_G}(V), \quad \forall V \in \text{Rep}_R(G).$$

Th. 2. R noeth. $\mathbb{Z}_\ell$ -alg, any f.g. sm. G -rep/ R is $\mathcal{Z}_R$ -finite, i.e.

1) $\text{Im}(\mathcal{Z}_R(G) \rightarrow \text{End}_{R_G}(V))$ is f.g. R -alg.

2) V is admissible over $\mathcal{Z}_R(G)$.

$(V^H \text{ f.g. } \mathcal{Z}_R(G)\text{-module, } \forall H \subset G \text{ open cpt})$

Th. 1 $\forall R$ noeth $\mathbb{Z}_\ell$ -alg, $\forall H$

1) $R[H \backslash G / H]$ is f.g. over its centre

2) centre is f.g. (comm.) R -algebra

Proof Thm 2

Reduction 1: $R = \mathbb{Z}_\ell[\sqrt{q}]$ suffices.

Pf of Red 1 (sketch) Easy for Th. 1.

$$R^1 \otimes_R \mathcal{H}_R = \mathcal{H}_{R^1}.$$

$$\mathbb{Z}_\ell \rightarrow \mathbb{Z}_\ell[\sqrt{q}]$$

know finiteness results for $\mathbb{Z}_\ell \Rightarrow$ same for R . is faithfully flat.

So finiteness over $\mathbb{Z}_\ell[\sqrt{q}] \Rightarrow$ over $\mathbb{Z}_\ell$ "□" R1.

Red 2 Reduce to $V = i_p(W)$
 $\hookrightarrow$ cusp. rep for M , $P = MU$
 ℓ -tor. free

Pf - Obs: Prop. of being $\mathcal{F}$ -finite is closed under subobjs and quotients.

- $\text{Rep}_R(G)$ is gen. by $R[G/H]$ where H cpt opens

$\nearrow$ (H pro-p).
 projective

$$P \in R^X, \quad \text{Hom}_G(R[G/H], V) = V^H \quad \text{right-exact}$$

- Claim. Any f.g. proj. RG -mod. Q embeds into a finite sum

$$\bigoplus_P i_p(W_P)$$

$\hookrightarrow$ cusp. f.g. ℓ -tor. free

Pf Claim. $Q \otimes \overline{\mathbb{Q}_\ell} \hookrightarrow \bigoplus_P i_p(\pi \otimes \overline{\mathbb{Q}_\ell}[M/M^0])$
 $\hookrightarrow$ cusp. M -ups.

$M^0 =$ Subgp gen. by compact subgps

Obs: up to unram. twists, we may assume each π has integral central char.

Subclaim: If π has integral central char. then it admits M -invariant lattice.

Know how to prove subclaim if $\pi \simeq \text{cind}_J^M P$.

(need char. assumptions)

$J \subset M$ open, cpt/centre, P finite dim'l

pt. $E|O_E$ finite, $\omega_\pi: \mathbb{Z}_M \rightarrow O_E^\times$ is also central char. of $p \hookrightarrow \pi|_J$.

p f.d. $\exists J_1 \subset J$ open normal acts trivially

$$1 \rightarrow \frac{J \cap \mathbb{Z}_M}{J_1 \cap \mathbb{Z}_M} \rightarrow \frac{J}{J_1} \rightarrow \frac{J}{(J \cap \mathbb{Z}_M)J_1} \rightarrow 1$$

$\uparrow$ $\uparrow$
 A B
 central pick set of lifts finite

Take any L inside p , average over these lifts.

A preserves all lattices.

L' is stable under J

$$\Lambda_\pi = \text{cind}_J^M(L')$$

M -stable lattice inside π subclaim " $\square$ "

Pick $E|O_E$ s.t.

$$Q \otimes E \hookrightarrow \bigoplus_p i_p(\dots) \text{ is defined.}$$

$$\varphi_{O_E}: Q \otimes O_E \hookrightarrow \bigoplus_p i_p(\Lambda_\pi \otimes O_E [M/M^0])$$

(since Q is b.g. proj., up to scaling by some power of l .
 we get an embedding like this.

$$\text{Hom}(Q \otimes O_E, \bigoplus_p \dots / O_E) \otimes_{O_E} E = \text{Hom}(Q_E, \dots)$$

Why is this map still injective?

$$\ker \varphi_{O_E} \text{ is torsion-free, } \otimes_{O_E} E = 0, \text{ so } \ker \varphi_{O_E} = 0.$$

each $\Lambda_\pi \otimes \mathcal{O}_E [M/M^\circ]$ ℓ -torsion free, cuspidal f.g.

$$\text{cind}_{M^\circ}^1 ((\Lambda_\pi)_{M^\circ})$$

claim $\square$

Red 2 $\square$

Assume $V = \overbrace{i^*p(W)}^{M \text{ rep}}$ as above.

Lemma (Dat '09) W cuspidal f.g. $\Rightarrow$ admissible over $R[\mathbb{Z}_M]$
 $\Rightarrow$ admissible over $\widehat{\mathbb{Z}_R}(\mathbb{Z}_M)$

Pr sketch Pick H pro- p open subgroup.

W f.g. $\Rightarrow W$ is gen. by W^H .

$$e_H \mathcal{H}_R(M) e_H \simeq W^H$$

cuspidal $\Rightarrow g \mapsto e_H g v$ have compact support / $\mathbb{Z}_M$ for any $v \in W^H$

$\Rightarrow e_H \mathcal{H}_R(M) e_H \langle v \rangle$ f.t. over $R[\mathbb{Z}_M]$.

W^H f.g. over $e_H \mathcal{H}_R(M) e_H$

$\Rightarrow W^H$ f.g. over $R[\mathbb{Z}_M]$.

$\square$ Lemma.

LLC: $\text{Irr}_k(G) \longrightarrow \{L\text{-parameters}\} / \hat{\mathbb{Z}}$

finite fibres, surj. $\Rightarrow$ s.s. L -par.

- loc. to obvious duality for for
- comp. twisting, central character
- contragredients, β - $\mathbb{Z}$ duality
- comp. w/ isog. of groups

- comp. w/ products, Weil restriction
- comp. w/ parabolic ind.

F-S constructs a correspondence in style of V. Lafforgue.

unram. Hecke $\subset$ $\overset{\text{Exc}}{\text{"Excursion"}}$ operators $\rightarrow \mathbb{C}^{\text{cusp}}$ some space of cusp. forms.

Smaller generalized eigenspace decomp.

χ char. of Exc. $\longleftrightarrow$ L-parameter.

char. by: $\chi \left(\underset{\substack{\uparrow \\ \text{Exc}}}{S_{I, t, (r_i)_i}} \right) = f(\sigma(r_i), \dots)$

so $\mathbb{C}^{\text{cusp}} = \bigoplus_{\sigma} \mathbb{C}_{\sigma}^{\text{cusp}}$ defines the L-packets.

$$\text{Exc}(W_F, \hat{G}) = \lim_e \text{Exc}(W_F / \underbrace{\mathfrak{p}_F^e}_{\text{discrete}}, \hat{G})$$

$$\text{For } W \text{ discrete, } \text{Exc}(W, \hat{G}) := \varinjlim_{n, F_n \rightarrow W} \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G}))^{\hat{G}}.$$

v.s.

$$\mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}}$$

There is a natural map $\text{Exc}(W, \hat{G})_{\mathbb{Z}_\ell} \rightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))_{\mathbb{Z}_\ell}^{\hat{G}}$

- universal homeomorphism

- iso. / $\mathcal{O}_\ell$ or if ℓ is good for G .

- away from ℓ -torsion, this is indep. of choice of discretization.

- same geom. pts. so enough info to pick out L-parameters

Th (F-S VII.4.1) $\exists$ nat'l map

$$FS_G: \text{Exc}(W_F, \hat{G})_R \rightarrow \mathcal{Z}_R(G)$$

1) comp. w/ parab. induction: composing w/ maps from $\mathcal{Z}_R(G)$

$W = W_F$
 $R = \mathbb{Z}_\ell'$

after Fed and
 fixing level e ,
 finite

$$\begin{array}{ccc} & \downarrow M\text{-rep } W_\theta & \\ & \mathcal{Z}_R(G) & \\ \text{Exc}(W_F, \hat{G})_R & \xrightarrow{\quad} & \text{End}_{R[G]}(\text{ip}(W_\theta)) \\ \downarrow & \cup & \uparrow \\ \text{Exc}(W_F, \hat{M})_R & \xrightarrow{\quad} & \text{End}_{RM}(W_\theta) \end{array}$$

2) comp. w/ central char.

$$\begin{array}{ccc} \text{Exc}(W_F, \hat{M})_R & \xrightarrow{\quad} & \mathcal{Z}_R(M) \\ \uparrow & & \uparrow \\ \text{Exc}(W_F, (\hat{M})_{ab})_R & \xrightarrow{\quad} & \mathcal{Z}_R(\mathbb{Z}_M) \\ \parallel & & \\ \hat{\mathbb{Z}}_M & & \end{array}$$

3) For tori $\mathbb{Z}_M$, $FS_{\mathbb{Z}_M}$ is isom.

4) W M -rep. f.g.

$\text{Exc}(W_F, \hat{M})_R \rightarrow \mathcal{Z}_R(M) \rightarrow \text{End}(W)$ factors through

$$\text{Exc}(W_F^\circ / P_F^e, \hat{M})_R \quad \text{for some } e$$

Where is LLC?

k alg. closed field

$\hat{G}$ -classes of ss L -par. / k

$\leftrightarrow$ cl. orbits in $Z'(W_F, \hat{G})/k$

$\leftrightarrow$ k -pts of $Z'(W_F, \hat{G})//\hat{G}$

$$\text{Exc}(W, \hat{G}) \approx \mathcal{O}(Z'(W_F, \hat{G}))^{\hat{G}} \rightarrow k$$

Cor. If k alg. closed, π is irrep, $\exists$ unique conj. class of ss. L -parameters

compatible w this map.

$\text{End}(\pi) = k$. $\text{Exc}(W_F, \hat{G}) \xrightarrow{FS} \mathcal{Z}_k(\hat{G}) \rightarrow \text{End}(\pi)$ defines L -parameter.

$\longleftarrow$

$$V = i_p(W), \quad R = \mathbb{Z}_\ell'.$$

W is $\mathcal{Z}_R(\mathbb{Z}_M)$ -adm.

$\Rightarrow$ by (2), (3), W is $\text{Exc}(W_F, \hat{M})_R$ -admissible.

by (4), W is adm. over $\text{Exc}(W_F^\circ/P_F^e, \hat{M})_R$ for some e .

Parabolic induction preserves admissibility over any given ring

so $i_p(W)$ is also $\text{Exc}(W_F^\circ/P_F^e, \hat{M})_R$ -admissible.

- after inverting ℓ , $\text{Exc}(\cdot)$ is reduced.

$\Rightarrow$ action on W or V factors through $\text{Exc}(\cdot)_{R, \text{red}}$ (nilradical ℓ -torsion)

Wenqing's finiteness:

V is adm. / this

$$\text{Exc}(w, \hat{u})_{R, \text{red}} \rightarrow \text{Exc}(w, \hat{u})_{R, \text{red}}$$

is finite

$$\Rightarrow V \text{ is } \text{Exc}(w_F, \hat{u})_{R, \text{red}} \text{ adm.}$$

$\uparrow$

$$\Rightarrow \text{Exc}(w_F, \hat{u})_R \text{ adm.}$$

$$\Rightarrow V \text{ is } \mathfrak{f}_R(u) \text{-adm.} \quad (2)$$

$$\text{Exc}(w_F, \hat{u})_R \rightarrow \mathfrak{f}_R(u) \rightarrow \text{End}_{R_H}(V)$$

$$\text{with } E_R \subset \mathbb{Z}_R \subset \text{End}_{R_H}(V)$$

for images

E_R is f.t. over $R \Rightarrow$ noetherian

If $\text{End}_{R_H}(V)$ is finite / E_R , done.

$$\text{End}_{R_H}(V) = \text{End}_{E_R H}(V)$$

know admissibility over E_R . So if we can find $H \subset G$ s.t.

$$\left(\begin{array}{l} \text{finite} \\ \text{happens whenever} \end{array} \right) \Leftrightarrow \frac{\text{finite} / E_R}{\text{finite} / E_R}$$

V^H generates V .

Don't know V f.g.

But we still claim we can do this.

Each V lives in bounded depth subcategory $\text{Rep}_R(\mathcal{A})_{\leq r}$.

$\downarrow$

has a single b.g. proj. generator

which is generated by its H -invariants $\square$

1. The first part of the document is a list of the names of the people who were present at the meeting.

2. The second part of the document is a list of the topics that were discussed.

3. The third part of the document is a list of the actions that were taken.